

Math 220: Mathematical Reasoning and Proof
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All Solutions

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Solutions 1: Introduction to course. First proof

Solutions to appear after the assignment is due.

1. Using the definitions of ‘divides’ and ‘even’, explain why 0 is even. Explain why 7 divides 0, but 0 does not divide 7.

Solution. Recall that an integer a is *even* if there exists an integer k such that $a = 2k$. Since $0 = 2 \cdot 0$ and 0 is an integer, it follows that 0 is even.

Recall also that, for integers a and b , we say that a *divides* b if there exists an integer k such that $b = ak$. Since $0 = 7 \cdot 0$, there exists an integer k , namely $k = 0$, such that $0 = 7k$. Therefore, $7 \mid 0$.

On the other hand, if 0 divided 7, then there would be an integer k such that $7 = 0 \cdot k$. But $0 \cdot k = 0$ for every integer k , so this would imply that $7 = 0$, which is false. Therefore, $0 \nmid 7$.

2. Suppose that n is an even integer, and let m be any integer. Prove that nm is even.

Solution. Since n is even, there exists an integer k such that $n = 2k$. Multiplying by m , we obtain

$$nm = (2k)m = 2(km).$$

Because k and m are integers, their product km is an integer. Thus nm is twice an integer, and we conclude that nm is even.

3. Suppose that n and m are odd integers. Prove that $n + m$ is an even integer.

Solution. Since n and m are odd, there exist integers a and b such that $n = 2a + 1$ and $m = 2b + 1$. Therefore,

$$n + m = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

Since a and b are integers, $a + b + 1$ is an integer. Thus $n + m$ is twice an integer, and we conclude that $n + m$ is even.

Solutions 2: Logic. “Direct” proofs

1. Write the negation of each of the following statements.

Solution.

- (a) All triangles are isosceles.

Negation: There exists a triangle that is not isosceles.

- (b) Every door in the building was locked.

Negation: There exists a door in the building that was not locked.

- (c) Some even numbers are multiples of three.

Negation: No even number is a multiple of three.

Equivalently: Every even number is not a multiple of three.

- (d) Every real number is less than 100.

Negation: There exists a real number that is greater than or equal to 100.

- (e) Every integer is positive or negative.

Negation: There exists an integer that is neither positive nor negative.

- (f) If f is a polynomial function, then f is continuous at 0.

Negation: f is a polynomial function and f is not continuous at 0.

- (g) If $x^2 > 0$, then $x > 0$.

Negation:

$$x^2 > 0 \quad \text{and} \quad x \leq 0.$$

- (h) There exists a $y \in \mathbf{R}$ such that $xy = 1$.

Negation: For every $y \in \mathbf{R}$,

$$xy \neq 1.$$

- (i) $(2 > 1)$ and $(\forall x, x^2 > 0)$.

Negation:

$$(2 \leq 1) \quad \text{or} \quad (\exists x \text{ such that } x^2 \leq 0).$$

- (j) $\forall \epsilon > 0, \exists \delta > 0$ such that if $|x| < \delta$, then $|f(x)| < \epsilon$.

Negation: There exists an $\epsilon > 0$ such that for every $\delta > 0$, $|x| < \delta$ and $|f(x)| \geq \epsilon$.

The following is also correct:

There exists an $\epsilon > 0$ such that for every $\delta > 0$, there exists an x such that

$$|x| < \delta \quad \text{and} \quad |f(x)| \geq \epsilon.$$

2. Write the converse, contrapositive, and negation of each of the following implications.

Solution.

- (a) If a quadrilateral is a rectangle, then it has two pairs of parallel sides.

Converse: If a quadrilateral has two pairs of parallel sides, then it is a rectangle.

Contrapositive: If a quadrilateral does not have two pairs of parallel sides, then it is not a rectangle.

Negation: There exists a quadrilateral that is a rectangle but does not have two pairs of parallel sides.

(b)

$$(P \wedge \neg Q) \Rightarrow R.$$

Converse:

$$R \Rightarrow (P \wedge \neg Q).$$

Contrapositive:

$$\neg R \Rightarrow \neg(P \wedge \neg Q).$$

Equivalently,

$$\neg R \Rightarrow (\neg P \vee Q).$$

Negation:

$$(P \wedge \neg Q) \wedge \neg R.$$

(c)

$$P \Rightarrow (R \Rightarrow \forall x, Q(x)).$$

Converse:

$$(R \Rightarrow \forall x, Q(x)) \Rightarrow P.$$

Contrapositive:

$$\neg(R \Rightarrow \forall x, Q(x)) \Rightarrow \neg P.$$

Equivalently,

$$(R \wedge \exists x \text{ such that } \neg Q(x)) \Rightarrow \neg P.$$

Negation:

$$P \wedge \neg(R \Rightarrow \forall x, Q(x)).$$

Equivalently,

$$P \wedge R \wedge \exists x \text{ such that } \neg Q(x).$$

3. Let P and Q be statements. Write the truth table for

(a) $(\neg P) \vee Q$

(b) $((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$

Solution.

(a)

P	Q	$\neg P$	$(\neg P) \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

(b)

P	Q	$\neg P$	$\neg P \Rightarrow Q$	$(\neg P \Rightarrow Q) \wedge \neg Q$	$((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$
T	T	F	T	F	T
T	F	F	T	T	T
F	T	T	T	F	T
F	F	T	F	F	T

4. Are the statements $(P \vee Q) \wedge R$ and $P \vee (Q \wedge R)$ equivalent? If so, give a proof. If not, explain why by giving a counterexample.

Solution.

They are not equivalent. For example, suppose $P = T$, $Q = F$, $R = F$. Then $(P \vee Q) \wedge R = (T \vee F) \wedge F = T \wedge F = F$, while $P \vee (Q \wedge R) = T \vee (F \wedge F) = T$. Thus the two statements can have different truth values, so they are not equivalent.

5. Let P and Q be statements.

- (a) Prove that $\neg(P \Rightarrow Q)$ is equivalent to $P \wedge \neg Q$.

Solution.

We can prove this using a truth table:

P	Q	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

The last two columns are identical. Therefore, $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$ are equivalent.

- (b) Prove that $\neg(P \Rightarrow Q)$ is *not* equivalent to $\neg P \wedge Q$.

Solution.

It is enough to find a choice of P and Q for which the two statements have different truth values.

Suppose $P = F$ and $Q = T$. Then $P \Rightarrow Q$ is true, so $\neg(P \Rightarrow Q) = F$. On the other hand, $\neg P \wedge Q = T \wedge T = T$. Thus the two statements are not equivalent.

- (c) Give an example of statements P and Q such that $\neg P \Rightarrow \neg Q$ is true and $\neg(P \Rightarrow Q)$ is false.

Solution.

For example, let

$$P = "2 > 1" \quad \text{and} \quad Q = "3 > 2".$$

Both P and Q are true.

Thus $\neg P$ is false, so $\neg P \Rightarrow \neg Q$ is true.

Also, $P \Rightarrow Q$ is true, and hence $\neg(P \Rightarrow Q)$ is false.

6. Suppose that n is an odd integer. Prove that n^2 is an odd integer.

Solution.

Since n is odd, there exists an integer k such that $n = 2k + 1$. Therefore,

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Since k is an integer, $2k^2 + 2k$ is an integer. Thus n^2 can be written in the form $2\ell + 1$ for some integer ℓ . Therefore n^2 is odd.

7. Prove that if n^2 is even, then n is even.

Solution.

We prove the statement by contraposition.

The contrapositive is:

If n is not even, then n^2 is not even.

Suppose that n is not even. Since n is an integer, n must therefore be odd. By the previous problem, if n is odd, then n^2 is odd. In particular, n^2 is not even.

Thus the contrapositive is true. Therefore the original statement is true: if n^2 is even, then n is even.

8. Prove that m is even if and only if $m^2 + 6$ is even.

Solution.

We prove both directions.

First suppose that m is even. Then there exists an integer k such that $m = 2k$. Therefore $m^2 + 6 = (2k)^2 + 6 = 4k^2 + 6 = 2(2k^2 + 3)$. Since $2k^2 + 3$ is an integer, $m^2 + 6$ is even.

Conversely, suppose that $m^2 + 6$ is even. Then there exists an integer k such that $m^2 + 6 = 2k$. Then $m^2 = 2k - 6 = 2(k - 6)$. Since $k - 6$ is an integer, m^2 is even. By the previous homework problem, we conclude that m is even.

Solutions 3: More “Direct” proofs; proof by cases

1. Prove that for all $a \in \mathbb{Z}$ and for $n \in \mathbb{Z}_{\geq 0}$, $a - 1$ divides $a^n - 1$.

Solution. If $n = 0$, then $a^n - 1 = 1 - 1 = 0 = (a - 1) \cdot 0$, so $a - 1 \mid a^n - 1$.

Now suppose $n \geq 1$. We use the factorization

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1).$$

The expression in the second set of parentheses is an integer, so by the definition of divisibility, $a - 1 \mid a^n - 1$.

2. Suppose that $a \mid b$. Prove that for all $n \in \mathbb{Z}_{>0}$, $a^n \mid b^n$.

Solution. Since $a \mid b$, there is an integer k such that $b = ak$. Raising both sides to the n th power gives

$$b^n = (ak)^n = a^n k^n.$$

Since k^n is an integer, this shows that $a^n \mid b^n$.

3. Suppose that there exists an integer $n \in \mathbb{Z}_{>0}$ such that $a \mid b^n$. Is it true that $a \mid b$? Prove or disprove your answer.

Solution. The statement is **false**. For example, take $a = 4$, $b = 2$, and $n = 2$. Then $b^n = 2^2 = 4$, so $4 \mid 2^2$. However, $4 \nmid 2$. Thus it is possible for $a \mid b^n$ even though $a \nmid b$.

4. Prove that for all integers n , n and $n + 1$ have no common divisors other than ± 1 .

Solution. Suppose that d is a common divisor of n and $n + 1$. Then there are integers r and s such that $n = dr$ and $n + 1 = ds$. Subtracting these equations gives $1 = (n + 1) - n = d(s - r)$. Thus $d \mid 1$. The only integer divisors of 1 are 1 and -1 , so $d = \pm 1$. Therefore n and $n + 1$ have no common divisors other than ± 1 .

5. Prove that the product of three consecutive integers is divisible by 6.

Solution. Let the three consecutive integers be n , $n + 1$, and $n + 2$. Among any two consecutive integers, one is even. Therefore at least one of these three integers is even, so $2 \mid n(n + 1)(n + 2)$.

Also, every integer has remainder 0, 1, or 2 when divided by 3. Thus among three consecutive integers, exactly one is divisible by 3. Hence $3 \mid n(n + 1)(n + 2)$.

Since 2 and 3 are relatively prime, divisibility by both 2 and 3 implies divisibility by 6. Therefore $6 \mid n(n + 1)(n + 2)$.

6. Show that for all integers a and b , $a^2 b^2 (a^2 - b^2)$ is divisible by 12.

Solution. We will show separately that the expression is divisible by 4 and by 3.

First consider divisibility by 4. If a is even, then a^2 is divisible by 4, so the entire product is divisible by 4. The same argument works if b is even.

The only remaining case is when both a and b are odd. Then both $a - b$ and $a + b$ are even. Since

$$a^2 - b^2 = (a - b)(a + b),$$

the factor $a^2 - b^2$ is divisible by 4. Thus in every case, $4 \mid a^2 b^2 (a^2 - b^2)$.

Now consider divisibility by 3. If $3 \mid a$ or $3 \mid b$, then $3 \mid a^2 b^2$, so the whole expression is divisible by 3.

Suppose instead that neither a nor b is divisible by 3. Then each of a and b has the form $3k + 1$ or $3k - 1$ for some integer k . In either case, its square has remainder 1 upon division by 3. Thus we can write $a^2 = 3r + 1$ and $b^2 = 3s + 1$ for some integers r, s . Therefore

$$a^2 - b^2 = (3r + 1) - (3s + 1) = 3(r - s),$$

so $3 \mid a^2 - b^2$. Hence in every case, $3 \mid a^2 b^2 (a^2 - b^2)$.

Since 3 and 4 are relatively prime, divisibility by both 3 and 4 implies divisibility by 12. Therefore $12 \mid a^2 b^2 (a^2 - b^2)$.

7. Finish the following proofs from class.

- (a) In class, when we proved that $|xy| = |x| |y|$, we only did the cases when $x \geq 0$. Prove that $|xy| = |x| |y|$ when $x < 0$.

Solution. Assume $x < 0$. There are two cases.

If $y \geq 0$, then $xy \leq 0$. Therefore $|xy| = -xy = (-x)y = |x| |y|$.

If $y < 0$, then $xy > 0$. Therefore $|xy| = xy = (-x)(-y) = |x| |y|$.

Thus in both cases, $|xy| = |x| |y|$.

- (b) In class, when we proved that $|x + y| \leq |x| + |y|$, we only did the cases when $x, y \geq 0$, when $x, y < 0$, and when $x \geq 0, y < 0$. Prove that $|x + y| \leq |x| + |y|$ when $x < 0$ and $y \geq 0$.

Solution. Assume $x < 0$ and $y \geq 0$. Then $|x| = -x$ and $|y| = y$. There are two cases depending on the sign of $x + y$.

If $x + y \geq 0$, then $|x + y| = x + y$. Since $x < 0$, we have $x \leq -x$, and so

$$|x + y| = x + y \leq -x + y = |x| + |y|.$$

If $x + y < 0$, then $|x + y| = -(x + y) = -x - y$. Since $y \geq 0$, we have $-y \leq y$, and therefore

$$|x + y| = -x - y \leq -x + y = |x| + |y|.$$

Thus in either case, $|x + y| \leq |x| + |y|$.

8. Consider the following statements about inequalities.

- (a) Prove that $|a - b| < c$ if and only if $b - c < a < b + c$.

Solution. We prove both directions.

First suppose that $|a - b| < c$. Since $|a - b| \geq 0$, this implies $c > 0$.

If $a - b \geq 0$, then $|a - b| = a - b < c$, so $a < b + c$. Also $a \geq b$, and since $c > 0$, we have $a \geq b > b - c$. Thus $b - c < a < b + c$.

If $a - b < 0$, then $|a - b| = b - a < c$, so $b - c < a$. Also $a < b$, and since $c > 0$, we have $a < b < b + c$. Again, $b - c < a < b + c$.

Conversely, suppose that $b - c < a < b + c$. Because a number a lies strictly between $b - c$ and $b + c$, we must have $c > 0$.

If $a - b \geq 0$, then $|a - b| = a - b$. From $a < b + c$, we get $a - b < c$, and hence $|a - b| < c$.

If $a - b < 0$, then $|a - b| = b - a$. From $b - c < a$, we get $b - a < c$, and hence $|a - b| < c$.

Therefore $|a - b| < c$ if and only if $b - c < a < b + c$.

- (b) Prove or disprove: if $a - b < c$, then $|a - b| < c$.

Solution. This statement is **false**. Take $a = 0$, $b = 2$, and $c = 1$. Then $a - b = -2 < 1 = c$, so the hypothesis is true. But $|a - b| = |-2| = 2 \not< 1$. Thus the conclusion is false.

- (c) Prove or disprove: if $|a - b| < c$, then $a - b < c$.

Solution. This statement is **true**. By part (a), $|a - b| < c$ implies $b - c < a < b + c$. In particular, $a < b + c$. Subtracting b from both sides gives $a - b < c$.

- (d) Prove that $||x| - |y|| \leq |x + y|$.

Solution. By the triangle inequality,

$$|x| = |(x + y) - y| \leq |x + y| + |y|.$$

Subtracting $|y|$ gives $|x| - |y| \leq |x + y|$. Similarly,

$$|y| = |(x + y) - x| \leq |x + y| + |x|,$$

so $|y| - |x| \leq |x + y|$.

Now there are two cases. If $|x| - |y| \geq 0$, then $||x| - |y|| = |x| - |y| \leq |x + y|$. If $|x| - |y| < 0$, then $||x| - |y|| = |y| - |x| \leq |x + y|$.

Therefore $\boxed{||x| - |y|| \leq |x + y|}$.

Solutions 4: Proof by contradiction

1. Prove that there do not exist integers a and b such that $21a + 30b = 1$.

Solution. Suppose, for contradiction, that there were integers a and b satisfying $21a + 30b = 1$. Both $21a$ and $30b$ are divisible by 3, so their sum is divisible by 3. Thus $3 \mid (21a + 30b)$. But $21a + 30b = 1$, so this would imply that $3 \mid 1$, which is impossible.

Therefore there are no integers a, b such that $21a + 30b = 1$.

2. Find all positive integers n such that $n^2 - 1$ is prime. Prove that your answer is correct.

Solution. We factor $n^2 - 1 = (n - 1)(n + 1)$.

If $n = 1$, then $n^2 - 1 = 0$, which is not prime. If $n = 2$, then $n^2 - 1 = 4 - 1 = 3$, which is prime.

Now suppose that $n \geq 3$. Then $n - 1 \geq 2$ and $n + 1 \geq 4$. Thus $n^2 - 1$ is a product of two integers greater than 1, so it is not prime.

Therefore the only positive integer n for which $n^2 - 1$ is prime is $\boxed{n = 2}$.

3. Prove that if x is an integer, then $x^2 + 2$ is not divisible by 4.

Solution. There are two cases.

If x is even, then $x = 2k$ for some integer k . Hence

$$x^2 + 2 = (2k)^2 + 2 = 4k^2 + 2.$$

If this were divisible by 4, then subtracting $4k^2$ would imply that 2 is divisible by 4, which is false.

If x is odd, then $x = 2k + 1$ for some integer k . Then

$$x^2 + 2 = (2k + 1)^2 + 2 = 4k^2 + 4k + 3 = 4(k^2 + k) + 3.$$

If this were divisible by 4, then subtracting $4(k^2 + k)$ would imply that 3 is divisible by 4, which is false.

Thus in either case, $\boxed{4 \nmid x^2 + 2}$.

4. Prove that there are no integer solutions to the equation $x^2 - y^2 = 2$.

Solution. Suppose, for contradiction, that integers x and y satisfy $x^2 - y^2 = 2$.

If x and y have the same parity, then either both are even or both are odd. If both are even, then both x^2 and y^2 are divisible by 4, so $x^2 - y^2$ is divisible by 4 and cannot equal 2.

If both are odd, write $x = 2r + 1$ and $y = 2s + 1$. Then

$$x^2 - y^2 = (4r^2 + 4r + 1) - (4s^2 + 4s + 1),$$

which is divisible by 4, so it cannot equal 2.

If x and y have different parity, then one of x^2, y^2 is even and the other is odd. Therefore $x^2 - y^2$ is odd, so it cannot equal 2.

All possibilities lead to a contradiction. Therefore there are no integer solutions.

5. Let a, b, c be integers satisfying $a^2 + b^2 = c^2$. Prove that at least one of a, b , or c is even.

Solution. Suppose, for contradiction, that none of a, b, c is even. Then all three are odd.

The square of an odd integer is odd, so a^2, b^2 , and c^2 are all odd. But $a^2 + b^2$ is the sum of two odd integers, and is therefore even, while c^2 is odd. Thus $a^2 + b^2 \neq c^2$, contradicting the hypothesis.

Therefore at least one of a, b, c must be even.

6. Suppose that x is a real number such that $0 \leq x \leq \pi/2$. Prove that $\sin x + \cos x \geq 1$.

Solution. Because $0 \leq x \leq \pi/2$, both $\sin x$ and $\cos x$ are nonnegative, so $\sin x + \cos x \geq 0$.

Now

$$\begin{aligned} (\sin x + \cos x)^2 &= (\sin x)^2 + 2 \sin x \cos x + (\cos x)^2 \\ &= 1 + 2 \sin x \cos x. \end{aligned}$$

Since $\sin x \geq 0$ and $\cos x \geq 0$, we have $2 \sin x \cos x \geq 0$. Hence $(\sin x + \cos x)^2 \geq 1$. Since $\sin x + \cos x$ is nonnegative, it follows that $\boxed{\sin x + \cos x \geq 1}$.

7. Prove that $2^{1/3}$ is irrational.

Solution. Suppose, for contradiction, that $2^{1/3}$ is rational. Then we can write $2^{1/3} = p/q$, where p and q are integers, $q \neq 0$, and p and q have no common divisor other than ± 1 .

Cubing both sides gives

$$2 = \frac{p^3}{q^3},$$

so $p^3 = 2q^3$. Thus p^3 is even. If p were odd, then p^3 would be odd, so p must be even. Write $p = 2k$. Substituting gives

$$(2k)^3 = 2q^3,$$

so $8k^3 = 2q^3$, and therefore $q^3 = 4k^3$. Thus q^3 is even, so q is even as well.

Therefore both p and q are even, so they have a common divisor 2. This contradicts our choice of p/q in lowest terms.

Hence $\boxed{2^{1/3} \text{ is irrational.}}$

8. Suppose that a and n are integers that are both at least 2, and suppose that $a^n - 1$ is prime.

- (a) Prove that $a = 2$.

Solution. We use the factorization

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1).$$

Because $a \geq 2$ and $n \geq 2$, the second factor is greater than 1.

Since $a^n - 1$ is prime, it cannot be written as a product of two positive integers that are both greater than 1. Therefore the first factor must equal 1, so $a - 1 = 1$. Hence $\boxed{a = 2}$.

- (b) Prove that n is prime.

Solution. Suppose, for contradiction, that n is not prime. Since $n \geq 2$, this means that n is composite, so $n = rs$ for some integers $r, s > 1$.

Then

$$a^n - 1 = a^{rs} - 1 = (a^r)^s - 1 = (a^r - 1)(a^{r(s-1)} + a^{r(s-2)} + \cdots + a^r + 1).$$

Because $a \geq 2$ and $r, s > 1$, both factors are greater than 1. Thus $a^n - 1$ is composite, contradicting the assumption that it is prime.

Therefore $\boxed{n \text{ is prime.}}$